

1 Introduction

We study the interpolated stochastic heat and wave equation (ISHWE) for spatial dimension on $\mathbb{R}^d$,

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = u_0 & b \in (0, 1] \\ u(0, \cdot) = u_0, \quad \partial_t u(0, \cdot) = v_0 & b \in (1, 2). \end{cases} \quad (1.1) \quad \boxed{\text{E:SPDE}}$$

2 Preliminaries

For any $t \in [0, T]$ and $x \in \mathbb{R}^d$, we write (assuming $\beta \in (1, 2)$)

$$u(t, x) = J_0(t, x) + I(t, x)$$

where

$$\begin{aligned} J_0(t, x) &= \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1] \\ [Z(t, \cdot) * v_0](x) + \frac{\partial}{\partial t}[Z(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases} \\ I(t, x) &= \int_0^t \int_{\mathbb{R}^d} Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy). \end{aligned}$$

To establish the existence and uniqueness of random field solutions to (1.1), we will need to assume Dalang's condition:

$$\int_0^s t ds \int_{\mathbb{R}^d} dy |Y(s, y)|^2 < \infty, \quad \text{for all } t > 0, \quad (2.1) \quad \boxed{\text{C:Dalang}}$$

which is equivalent to

$$d < 2\alpha + \frac{\alpha}{\beta} \min\{2\gamma - 1, 0\} =: \Theta. \quad (2.2) \quad \boxed{\text{E:DalangEqu}}$$

3 1-d Case

Lemma 3.1. [2, Theorem 4.1 and Lemma 5.5] *Assume that $d < 2\alpha$, $\beta \in (0, 2)$, and $\gamma \geq 0$. Then*

$$\int_{\mathbb{R}^d} Y(t, x) dx = t^{\beta+\gamma-1}$$

and

$$\int_{\mathbb{R}^d} Y^2(t, x) dx = C_{\#} t^{2(\beta+\gamma-1)-d\beta/\alpha},$$

for all $t > 0$, where

$$C_{\#} := \frac{2}{\Gamma(d/2)(4\pi)^{d/2}} \int_0^\infty u^{d-1} E_{\beta, \beta+\gamma}^2(-u^\alpha) du.$$

Now further assume that $\beta \in (1, 2)$, then

$$\int_{\mathbb{R}} Z(t, x) = t^{[\beta]-1} = t.$$

Remark 3.2. When $\alpha = \beta = 2$, $\gamma = 0$ and $d = 1$ then Lemma 3.1 gives us that $\int_{\mathbb{R}} G^2(t, x) dx = t/2$ which coincides with the $L^2(\mathbb{R})$ norm of the wave kernel.

In the proof of the next proposition, we will use the following two facts:

$$\sup_{t \geq 0} t^k e^{-at} = k^k (ea)^{-k} \text{ for } k \in \mathbb{N} \text{ and } \sup_{t \geq 0} \int_0^t s e^{-as} ds = a^{-2} \text{ for } a > 0. \quad (3.1) \quad \boxed{\text{E:SUPidenti}}$$

Proposition 3.3. [3, Proposition 3.2] *Let u_0 and v_0 be Borel functions satisfying $\|u_0\|_\infty + \|v_0\|_\infty < \infty$. Then there exists a universal constant $K := K(\alpha, \beta, \gamma) > 0$ such that for any*

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-1)+\alpha(2\gamma-1))}.$$

and

$$p \in \left[2, \frac{a^{(\beta(2\alpha-1)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2} \right].$$

we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (3.2) \quad \text{E:Nbddd}$$

where

$$\mathcal{T}_0 = \begin{cases} ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|u_0\|_\infty & \beta \in (0, 1] \\ ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \right]^p, \quad t \in [0, T]. \quad (3.3) \quad \text{E:Pnormtbdd}$$

Proof. First we fix an $a > 0$ and $p \in [2, \infty)$. Using Lemma 3.1 we see that,

$$\begin{aligned} |J(t, x)| &= \begin{cases} |(Z(t, \cdot) * u_0)(x)| & \beta \in (0, 1] \\ |(Z(t, \cdot) * v_0)(x) + \frac{\partial}{\partial t}(Z(t, \cdot) * u_0)(x)| & \beta \in (1, 2) \end{cases} \\ &\leq \begin{cases} t^{[\beta]-1} \|u_0\|_\infty & \beta \in (0, 1] \\ t^{[\beta]-1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \end{aligned}$$

Now using (3.1), we see that

$$\begin{aligned} N_{a,p}(J) &\leq \begin{cases} \sup_{t \geq 0} \sup_{x \in \mathbb{R}} t^{[\beta]-1} \|u_0\|_\infty e^{-at} & \beta \in (0, 1] \\ \sup_{t \geq 0} \sup_{x \in \mathbb{R}} (t^{[\beta]-1} \|v_0\|_\infty + \|u_0\|_\infty) e^{-at} & \beta \in (1, 2) \end{cases} \\ &= \begin{cases} ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|u_0\|_\infty & \beta \in (0, 1] \\ ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases} \quad (3.4) \quad \text{E:Jbdd} \end{aligned}$$

Now we find an upper bound for $N_{a,p}(I)$. By simultaneously applying the Burkholder-Davies-Gundy inequality and triangle inequality, we get that

$$\begin{aligned} \|I(t, x)\|_p^2 &\leq 4p \left\| \int_0^t ds \int_{\mathbb{R}} dy Y^2(t-s, x-y) \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^t ds \int_{\mathbb{R}} dy Y^2(t-s, x-y) \|\sigma^2(u(s, y))\|_{p/2}. \end{aligned}$$

Now by applying Lemma 3.1 and the Lipschitz property of σ we get that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \int_0^t (t-s)^{2(\beta+\gamma-1)-\beta/\alpha} ds + 8pL^2(\sigma)N_{a,p}^2(u) \int_0^t ds \int_{\mathbb{R}} dy e^{2as} Y^2(t-s, x-y).$$

Now under Dalang's condition [2], $(1/\alpha - 2)\beta - 2\gamma < -1$, the first integral above can be calculated as

$$\int_0^t (t-s)^{2(\beta+\gamma-1)-\beta/\alpha} ds = \frac{\alpha t^{(2-1/\alpha)\beta+2\gamma-1}}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta}.$$

By applying Lemma 3.1, we can calculate the second integral as

$$\int_0^t ds \int_{\mathbb{R}} dy e^{2as} Y^2(t-s, x-y) = C_{\#} \int_0^t ds e^{2as} (t-s)^{2(\beta+\gamma-1)-\beta/\alpha}.$$

Putting this together gives us that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma) C_{\#} \frac{\alpha t^{(2-1/\alpha)\beta+2\gamma-1}}{2\alpha\beta+2\alpha\gamma-\alpha-\beta} + 8pL^2(\sigma) N_{a,p}^2(u) C_{\#} \int_0^t ds e^{2as} (t-s)^{2(\beta+\gamma-1)-\beta/\alpha}.$$

Now by multiplying both sides by e^{-2at} , we see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma) C_{\#} \alpha}{2\alpha\beta+2\alpha\gamma-\alpha-\beta} t^{(2-1/\alpha)\beta+2\gamma-1} e^{-2at} \\ &\quad + 8pL^2(\sigma) N_{a,p}^2(u) C_{\#} \int_0^t ds e^{-2a(t-s)} (t-s)^{2(\beta+\gamma-1)-\beta/\alpha}. \end{aligned}$$

We now do a change of variables and again recall the definition of the Gamma function to see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma) C_{\#} \alpha}{2\alpha\beta+2\alpha\gamma-\alpha-\beta} t^{(2-1/\alpha)\beta+2\gamma-1} e^{-2at} \\ &\quad + 8pL^2(\sigma) N_{a,p}^2(u) C_{\#} \int_0^\infty du e^{-2a(u)} (u)^{2(\beta+\gamma-1)-\beta/\alpha} \\ &= \frac{8pc^2(\sigma) C_{\#} \alpha}{2\alpha\beta+2\alpha\gamma-\alpha-\beta} t^{(2-1/\alpha)\beta+2\gamma-1} e^{-2at} \\ &\quad + 8pL^2(\sigma) N_{a,p}^2(u) C_{\#} (2a)^{1+(1/\alpha-2)\beta-2\gamma} \Gamma[(2-1/\alpha)\beta-1+2\gamma] \end{aligned}$$

where the last line is true if Dalang's condition [2] holds. If we now apply (3.1) and also recall that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, then we see that

$$\begin{aligned} N_{a,p}(I) &\leq c(\sigma) \sqrt{\frac{8pC_{\#}\alpha}{2\alpha\beta+2\alpha\gamma-\alpha-\beta}} [(2-1/\alpha)\beta/2 + \gamma - 1/2]^{(2-1/\alpha)\beta/2 + \gamma - 1/2} (ea)^{1/2 - \gamma - (2-1/\alpha)\beta/2} \\ &\quad + L(\sigma) N_{a,p}(u) \sqrt{8pC_{\#}(2a)^{1+(1/\alpha-2)\beta-2\gamma} \Gamma[(2-1/\alpha)\beta-1+2\gamma]}. \end{aligned} \quad (3.5) \quad \boxed{\text{E:SoleMil3.}}$$

To simplify things, we now let

$$\begin{aligned} K := \max \left\{ \sqrt{\frac{8C_{\#}\alpha}{2\alpha\beta+2\alpha\gamma-\alpha-\beta}} \left(\frac{(2-1/\alpha)\beta/2 + \gamma - 1/2}{e} \right)^{(2-1/\alpha)\beta/2 + \gamma - 1/2}, \right. \\ \left. \left(\frac{1}{2} \right)^{(2-1/\alpha)\beta/2 + \gamma - 1/2} \sqrt{8C_{\#}\Gamma[(2-1/\alpha)\beta-1+2\gamma]} \right\} \end{aligned}$$

then,

$$N_{a,p}(I) \leq \frac{K\sqrt{p}}{a^{(2-1/\alpha)\beta/2 + \gamma - 1/2}} (c(\sigma) + L(\sigma) N_{a,p}(u)). \quad (3.6) \quad \boxed{\text{E:Ibdd}}$$

Now by combining (3.4) and (3.6), we see that

$$N_{a,p}(u) \leq \begin{cases} ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|u_0\|_\infty & \beta \in (0, 1] \\ ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases} \quad (3.7)$$

$$+ K \frac{\sqrt{p}c(\sigma)}{a^{(2-1/\alpha)\beta/2 + \gamma - 1/2}} + K \frac{\sqrt{p}L(\sigma)}{a^{(2-1/\alpha)\beta/2 + \gamma - 1/2}} N_{a,p}(u). \quad (3.8) \quad \boxed{\text{E:Nbddgen}}$$

It is important to note that if we now restrict ourselves to the case of $\alpha = \beta = 2$ and $\gamma = 0$, then (3.5) and (3.8) recover [3, (3.9) and (3.10)] respectively. Now choose $a > 0$ such that

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-1)+\alpha(2\gamma-1))}.$$

This implies that the following interval is nonempty:

$$\left[2, \frac{a^{(\beta(2\alpha-1)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2}\right].$$

In addition, for any p in this interval, we have that

$$\frac{\sqrt{p}L(\sigma)K}{a^{(2-1/\alpha)\beta/2+\gamma-1/2}} \leq \frac{1}{2},$$

and we also have

$$\frac{\sqrt{p}}{a^{(2-1/\alpha)\beta/2+\gamma-1/2}} \leq \frac{1}{2KL(\sigma)}.$$

This along with (3.8) implies that

$$\begin{aligned} N_{a,p}(u) &\leq 2 \begin{cases} (\lceil\beta\rceil - 1)^{\lceil\beta\rceil-1} (ea)^{1-\lceil\beta\rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil\beta\rceil - 1)^{\lceil\beta\rceil-1} (ea)^{1-\lceil\beta\rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \\ &\quad + \frac{c(\sigma)}{L(\sigma)} \end{aligned}$$

and this is precisely (3.2). Note that (3.3) follows immediately from (3.2). $\square$

We will need to recall the following result proved in [2] in order to prove Proposition 3.5 below.

Not quite correct, edit this later. When $1 < \beta < 2$ then we need $\rho < 2(\beta + \gamma) - 1 - \beta/\alpha$ [1, Proposition 2.2].

Proposition 3.4. [2, Proposition 5.4] *Suppose that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, and (2.2) holds. Then $Y(t, x) = Y_{\alpha, \beta, \gamma, d}(t, x)$ satisfies the following:*

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^\rho + |x-z|^\theta \right),$$

where $\rho = 2(\beta + \gamma) - 1 - \beta/\alpha$ and $0 < \theta < (\Theta - 1) \wedge 2$ and Θ is given in (2.2).

Proposition 3.5. *Suppose that $u_0 = 1$ and that $v_0 = 0$ so that $u(t, x) = 1 + I(t, x)$. Let $\rho = 2(\beta + \gamma) - 1 - \beta/\alpha$ and suppose that $0 < \theta < (\Theta - 1) \wedge 2$. Then,*

$$\frac{\|u(t, x) - u(r, z)\|_p}{(|t-r|^\rho + |x-z|^\theta)^{1/2}} \leq C(p, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)], \quad (3.9)$$

where

$$\mathcal{M}_1 = \sqrt{p}c(\sigma) \quad \text{and} \quad \mathcal{M}_2 = \sqrt{p}L(\sigma).$$

Moreover, for any

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-1)+\alpha(2\gamma-1))}.$$

and

$$p \in \left[2, \frac{a^{(\beta(2\alpha-1)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2}\right]$$

we have that

$$\frac{\|u(t, x) - u(r, z)\|_p}{(|t-r|^\rho + |x-z|^\theta)^{1/2}} \leq C(p, T) \left[\mathcal{M}_1 + \mathcal{M}_2 e^{aT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right) \right]. \quad (3.10)$$

Proof. Note that because of the initial conditions, we see that $\|u(t, x) - u(r, z)\|_p = \|I(t, x) - I(r, z)\|_p$. We now apply Burkholder-Davies-Gundy inequality along with the triangle inequality, as was done in the proof of Proposition 3.3, to see that

$$\begin{aligned}\|I(t, x) - I(r, z)\|_p^2 &\leq 4p \left\| \int_0^T ds \int_{\mathbb{R}} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^T ds \int_{\mathbb{R}} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \|\sigma^2(u(s, y))\|_{p/2}.\end{aligned}$$

Now by applying the Lipchitz condition on σ gives us that

$$\begin{aligned}\|I(t, x) - I(r, z)\|_p^2 &\leq 8p \int_0^T ds \int_{\mathbb{R}} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \left(c^2(\sigma) + L^2(\sigma) \|u(s, y)\|_p^2 \right) \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \int_0^T ds \int_{\mathbb{R}} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \exp(2as) N_{a,p}^2(u) \right] \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) \int_0^T ds \int_{\mathbb{R}} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right].\end{aligned}$$

We now apply Proposition 3.5 to see that

$$\begin{aligned}\|I(t, x) - I(r, z)\|_p^2 &\leq 8p \left[2c^2(\sigma) C(T) \left(|t-r|^\rho + |x-z|^\theta \right) \right. \\ &\quad \left. + 2L^2(\sigma) \exp(2aT) N_{a,p}^2(u) C(T) \left(|t-r|^\rho + |x-z|^\theta \right) \right].\end{aligned}$$

Now by taking the square root of both sides and applying the identity $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ yields

$$\|I(t, x) - I(r, z)\|_p \leq 4\sqrt{C(T)p} \left[c(\sigma) + L(\sigma) \exp(aT) N_{a,p}(u) \right] \left(|t-r|^\rho + |x-z|^\theta \right)^{1/2}.$$

This proves (3.9). It is clear that (3.10) follows directly from (3.9) and Proposition 3.3. $\square$

References

- [1] Chen, L and Hu, G. (2021). Holder regularity of the nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$. *arXiv:2105.00891*.
- [2] Chen, L., Hu, Y., and Nualart, D. (2019). Nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$. *Stochastic Process. Appl.* **129**, 5073–5112
- [3] Sanz-Sole, M. and Millet, A. (2021) *Global solutions to stochastic wave equations with superlinear coefficients*. Stochastic Processes and their Applications, Elsevier, 2021, 139, pp.175-211.